



Technical Service Bulletin

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New Intake Valve

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This Technical Service Bulletin introduces new intake valves for the QSK19 MCRS series construction engines (except as noted below). The new intake valve utilizes a nitride hardening process known as Sursulf. The benefits of Sursulf are:

- Increase surface hardness
- Increased fatigue strength
- Improved wear resistance.

Sursulf valves can **not** be ground. Grinding will remove a significant portion of the hardened layer.

Ground valves will be susceptible to "beat-in" wear and premature failure. It is **not** recommended to intermix old intake valve part numbers with the new Sursulf valve part numbers in the same cylinder head. Cylinder head combinations are acceptable on the same engine. Should there be a requirement to change the old intake valve, Part Number 3052820, it should be replaced with the new Sursulf valve, Part Number 4986317.

The new QSK19 MCRS Sursulf intake valve, Part Number 4986317, is a direct replacement for intake valve, Part Number 3052820.

The new valves have a darker appearance when compared to the old valves. The new valve can also be identified by the letter "O" forged onto the bottom center of the valve and by the part number laser-etched into the stem, above the collet groove. See Figure 1.

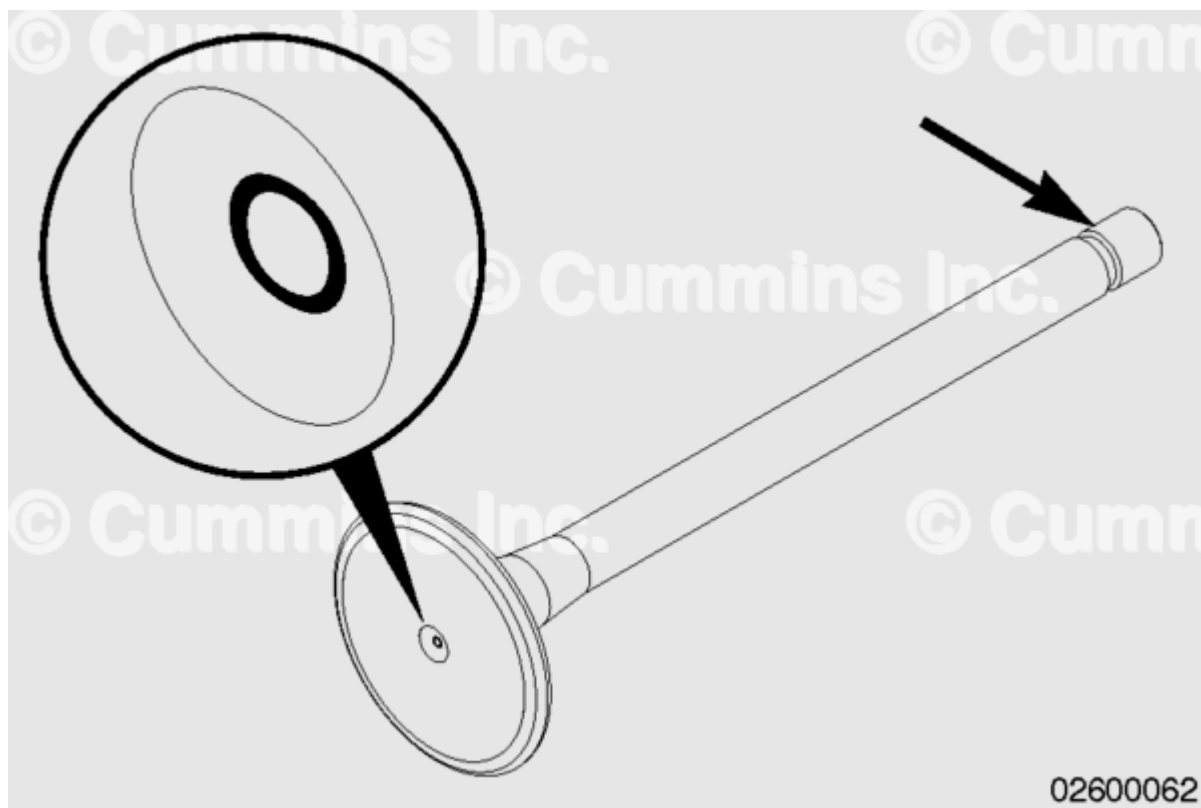


Figure 1: New QSK19 Intake Valve

The new intake valve is **only** available as a kit, with two new collets, Part Number 2881836.

NOTE: Engines manufactured at Cummins India Ltd. and **all** Marine Propulsion engines continue to use the old valve part numbers.